Database Design Exercise: Organizational Networking Platform

TABLES:

1) User

Table Structure

Column Name	Data Type	Constraints
user_id	integer	PK, Auto Increment, Not Null, Unique
user_name	char	Not Null
password	varchar(255)	Not Null
email	varchar(255)	Not Null, Unique
updated_at	varchar(255)	Not Null

Foreign Key Relationships

None

2) Organization

Table Structure

Column Name	Data Type	Constraints
organization_id	integer	PK, Auto Increment, Not Null, Unique
organization_name	char(1)	Not Null

Foreign Key Relationships

None

3) User_Organization

Table Structure

Column Name	Data Type	Constraints
user_id	integer	FK → user.user_id, Not Null
organization_id	integer	FK → organization.organization_id, Not Null
member_type	char(1)	Not Null

Foreign Key Relationships

- user_id → User(user_id)
- organization_id → Organization(organization_id)

4) Position

Table Structure

Column Name	Data Type	Constraints
position_id	integer	PK, Auto Increment, Not Null, Unique
organization_id	integer	FK → organization.organization_id, Not Null
title	char(1)	Not Null
is_vacant	char(1)	Not Null

Foreign Key Relationships

- organization_id → Organization(organization_id)

5) User_Request

Table Structure

Column Name	Data Type	Constraints
user_id	integer	FK → user.user_id, Not Null
organization_id	integer	FK → organization.organization_id, Not Null
position_id	integer	FK → position.position_id
request_type	char(1)	Not Null
status	char(1)	Not Null

Foreign Key Relationships

- user_id → User(user_id)
- organization_id → Organization(organization_id)
- position_id → Position(position_id)

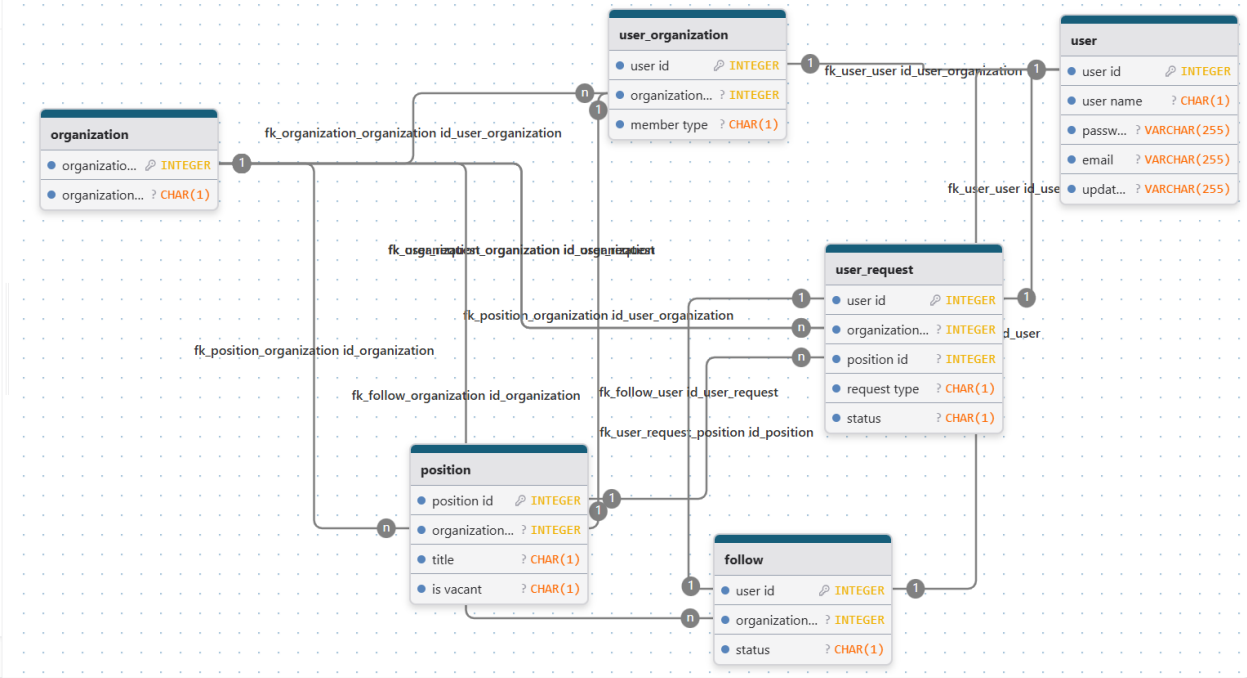
6) Follow

Table Structure

Column Name	Data Type	Constraints
user_id	integer	FK → user.user_id, Not Null
organization_id	integer	FK → organization.organization_id
status	char(1)	Not Null

Foreign Key Relationships

- user_id → User(user_id)
- organization_id → Organization(organization_id)



Relationship Table

Relationship Name	Cardinality
User – Organization	**M : N**
Organization – Position	**1 : M**
User – Position (via request)	**1 : M** (optional)
User – User (Follow)	**M : M**
User – User_Request	**1 : M**
Organization – User_Request	**1 : M**
Position – User_Request	**1 : M** (optional)